1.31 Differentiation: Introduction

Chapter 12 · Original slide 14 · Differentiating {x}^{n}

Thankfully, there's a quick way to differentiate terms of the form x^n (where n is a constant) with having to use first principles every time:

! If $y = ax^n$ then $\frac{dy}{dx} = nax^{n-1}$ (where a, n are constants)
i.e. multiply by the power and reduce the power by 1

Examples:

$y = x^5$

Power is 5, so multiply by 5 then reduce power by 5.

The power need not be an integer!
Remember to use $f'(x)$ not $\frac{dy}{dx}$

$f(x) = x^{\frac{1}{2}}$

$y = 2x^6$

Why would it be incorrect to say that $\mathbf{y=2^x}$ differentiates to $\mathbf{\frac{dy}{dx}=x2^{x-1}}$?
The rule only works when the base is x and the power is a constant. Neither is true here! Note that x^n is "a power of x" whereas 2^x is an exponential term (which you will encounter more in Chp14), and therefore differentiate differently. You will learn how to differentiate exponential terms in Year 2.

$f(x) = \frac{x}{x^4}$

$y = \sqrt{x^6}$

Chapter 12 · Original slide 15 · Test Your Understanding

1

$y = x^7$

2

$y = 3x^{10}$

3

$f(x) = \frac{x^{\frac{1}{2}}}{x^2}$

4

$y = ax^a$

5

$f(x) = \sqrt{49x^7}$

1.31 Differentiation: Introduction

Chapter 12 · Original slide 16 · Differentiating Multiple Terms

Differentiate $y = x^2 + 4x + 3$

First thing to note:

If $y = f(x) + g(x)$ then $\frac{dy}{dx} = f'(x) + g'(x)$

i.e. differentiate each term individually in a sum/subtraction.

$y = 4x = 4x^1$

Therefore applying the usual rule:

$\frac{dy}{dx} = 4x^0 = 4$

Alternatively, if you compare $y = 4x$ to $y = mx + c$, it's clear that the gradient is fixed and $m = 4$.

$y = 3 = 3x^0$

Therefore applying the usual rule:

$\frac{dy}{dx} = 0x^{-1} = 0$

Alternatively, if you sketch $y = 4$, the line is horizontal, so the gradient is 0.

Chapter 12 · Original slide 17 · Quickfire Questions

1

$y = 2x^2 - 3x$

2

$y = 4 - 9x^3$

3

$y = 5x + 1$

4

$y = ax$

(where a is a constant)

(where p is a constant)

5

$y = 6x - 3 + px^2$

1.31 Differentiation: Introduction

Chapter 12 · Original slide 18 · Harder Example

Let $f(x) = 4x^2 - 8x + 3$

a) Find the gradient of $y = f(x)$ at the point $\left(\frac{1}{2}, 0\right)$

b) Find the coordinates of the point on the graph of $y = f(x)$ where the gradient is 8.

c) Find the gradient of $y = f(x)$ at the points where the curve meets the line $y = 4x - 5$.

Remember that the 'gradient function' allows you to find the gradient for a particular value of x.

b

This example is important!

Previously you used a value of x to get the gradient $f'(x)$. This time we're doing the opposite: using a known gradient $f'(x)$ to get the value of x. We therefore substitute $f'(x)$ for 8.

c

Once you have your x, you need to work out y.

Ensure you use the correct equation!

Chapter 12 · Original slide 19 · Test Your Understanding

Let $f(x) = x^2 - 4x + 2$

a) Find the gradient of $y = f(x)$ at the point $(1, -1)$

b) Find the coordinates of the point on the graph of $y = f(x)$ where the gradient is 5.

c) Find the gradient of $y = f(x)$ at the points where the curve meets the line $y = 2 - x$.

1.31 Differentiation: Introduction

Chapter 12 · Original slide 20 · Exercise 12D – Pearson Pure Y1 page 265-266

5 Find the gradients of the curve $y = x^2 - 5x + 10$ at the points A and B where the curve meets the line $y = 4$.

Chapter 12 · Original slide 21 · Exercise 12D – Pearson Pure Y1 page 265-266

6 Find the gradients of the curve $y = 2x^2$ at the points C and D where the curve meets the line $y = x + 3$.

1.31 Differentiation: Introduction

Chapter 12 · Original slide 22 · Exercise 12D – Pearson Pure Y1 page 265-266

7 $f(x) = x^2 - 2x - 8$.
a Sketch the graph of $y = f(x)$.
b On the same set of axes, sketch the graph of $y = f'(x)$.
c Explain why the x-coordinate of the turning point of $y = f(x)$ is the same as the x-coordinate of the point where the graph of $y = f'(x)$ crosses the x-axis.

Chapter 12 · Original slide 23 · Differentiating Harder Expressions

If your expression isn't a sum of x^n terms, simply manipulate it until it is!

$y = \sqrt{x}$

2. Split up fractions.

3. Expand out brackets.

NOT $3x^{-1}$!!!

$y = \frac{1}{3x}$

1. Turn roots into powers:

$y = \frac{1}{\sqrt[3]{x}}$

$y = \frac{x^2+3}{\sqrt{x}}$

$y = x^2(x - 3)$

4. Beware of numbers in denominators!

1.31 Differentiation: Introduction

Chapter 12 · Original slide 24 · Test Your Understanding

Differentiate the following.

$y = \frac{1}{\sqrt{x}}$

$y = \frac{2+x^3}{x^2}$

$y = \frac{1+2x}{3x\sqrt{x}}$

Chapter 12 · Original slide 25 · Exercise 12E – Pearson Pure Y1 page 267-268

6 $\mathrm{f}(x) = \frac{12}{p\sqrt{x}} + x$, where p is a real constant and $x > 0$. Given that $\mathrm{f}'(2) = 3$, find p, giving your answer in the form $a\sqrt{2}$ where a is a rational number.

1.31 Differentiation: Introduction

Chapter 12 · Original slide 26 · Exercise 12E – Pearson Pure Y1 page 267-268

7 $\mathrm{f}(x) = (2 - x)^9$.
a Find the first 3 terms, in ascending powers of x, of the binomial expansion of $\mathrm{f}(x)$, giving each term in its simplest form.
b If x is small, so that x^2 and higher powers can be ignored, show that $\mathrm{f}'(x) \approx 9216x - 2304$.

Chapter 12 · Original slide 27 · Extension

[MAT 2013 1E]

The expression
$\frac{d^2}{dx^2}[(2x-1)^4(1-x)^5] + \frac{d}{dx}[(2x+1)^4(3x^2-2)^2]$
is a polynomial of degree:

9

8

7

less than 7

This just means “differentiate twice”. The ‘second derivative’ comes later in this chapter.

1.31 Differentiation: Introduction

Chapter 12 · Original slide 42 · Second Order Derivatives

When you differentiate once, the expression you get is known as the first derivative.

Unsurprisingly, when we differentiate a second time, the resulting expression is known as the second derivative. And so on…

Original Function

First Derivative

Second Derivative

Leibniz's

$\frac{dy}{dx} = 4x^3$

$\frac{d^2y}{dx^2} = 12x^2$

$y = x^4$

$y\prime = 4x^3$

$y\prime\prime = 12x^2$

Newton's

$\dot{y} = 4x^3$

$\ddot{y} = 12x^2$

$f(x) = x^4$

Lagrange's

$f\prime(x) = 4x^3$

$f\prime\prime(x) = 12x^2$

You can similarly have the third derivative ($\frac{d^3y}{dx^3}$), although this is no longer in the A Level syllabus. We'll see why might use the second derivative soon…

Chapter 12 · Original slide 43 · Just for your interest…

How does the notation $\frac{d^2y}{dx^2}$ work?

Why are the squareds where they are?

The $\frac{d}{dx}(\dots)$ notation is quite handy, because it behaves as a function and allows us to write the original expression and the derivative within a single equation:

$\frac{d}{dx}\left(x^3 + 1\right) = 3x^2$

Suppose that $y = x^3 + 1$.

Then when we write $\frac{dy}{dx}$, we're effectively doing $\frac{d(x^3+1)}{dx}$ (by substitution), although this would typically be written:

$\frac{d}{dx}\left(x^3 + 1\right)$

Therefore, if we wanted to differentiate y twice, we'd do:

$\frac{d}{dx}\left(\frac{d}{dx}(y)\right)$
$= \frac{d^2}{dx^2}(y)$
$= \frac{d^2y}{dx^2}$

1.31 Differentiation: Introduction

Chapter 12 · Original slide 44 · Examples

If $y = 3x^5 + \frac{4}{x^2}$, find $\frac{d^2y}{dx^2}$.

If $f(x) = 3\sqrt{x} + \frac{1}{2\sqrt{x}}$, find $f''(x)$.

Chapter 12 · Original slide 45 · Test Your Understanding

If $y = 5x^3 - \frac{x}{3\sqrt{x}}$, find $\frac{d^2y}{dx^2}$.

1.31 Differentiation: Introduction

Chapter 12 · Original slide 46 · Exercise 12H – Pearson Pure Y1 page 272

3 Given that $y = (2x - 3)^3$, find the value of x when $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$.

Chapter 12 · Original slide 47 · Exercise 12H – Pearson Pure Y1 page 272

4 $\mathrm{f}(x) = px^3 - 3px^2 + x^2 - 4$. When $x = 2$, $\mathrm{f}''(x) = -1$. Find the value of p.